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Protocol

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ProBac-seq, a bacterial single-cell RNA sequencing methodology using droplet microfluidics and large oligonucleotide probe sets

Prosenjit Samanta [1], Samuel F. Cooke[1], Ryan McNulty[2], Sahand Hormoz[3,4,5] & Adam Rosenthal [1]

Abstract

Methods that measure the transcriptomic state of thousands of individual cells have transformed our understanding of cellular heterogeneity in eukaryotic cells since their introduction in the past decade. While simple and accessible protocols and commercial products are now available for the processing of mammalian cells, these existing technologies are incompatible with use in bacterial samples for several fundamental reasons including the absence of polyadenylation on bacterial messenger RNA, the instability of bacterial transcripts and the incompatibility of bacterial cell morphology with existing methodologies. Recently, we developed ProBac sequencing (ProBac-seq), a method that overcomes these technical difficulties and provides high-quality single-cell gene expression data from thousands of bacterial cells by using messenger RNA-specific probes. Here we provide details for designing large oligonucleotide probe sets for an organism of choice, amplifying probe sets to produce sufficient quantities for repeated experiments, adding unique molecular indexes and poly-A tails to produce finalized probes, in situ probe hybridization and single-cell encapsulation and library preparation. This protocol, from the probe amplification to the library preparation, requires ~7 d to complete. ProBac-seq offers several advantages over other methods by capturing only the desired target sequences and avoiding nondesired transcripts, such as highly abundant ribosomal RNA, thus enriching for signal that better informs on cellular state. The use of multiple probes per gene can detect meaningful single-cell signals from cells expressing transcripts to a lesser degree or those grown in minimal media and other environmentally relevant conditions in which cells are less active. ProBac-seq is also compatible with other organisms that can be profiled by in situ hybridization techniques.

Key points

- This protocol describes ProBac sequencing, a bacterial single-cell RNA sequencing method using droplet microfluidics and large oligonucleotide probe sets.
- ProBac sequencing offers advantages over other methods. By capturing only the desired target sequences and avoiding nondesired transcripts such as highly abundant ribosomal RNA, it better informs on cellular state. Using multiple probes per gene allows meaningful single-cell signals to be detected from less active cells or those expressing transcripts to a lesser degree.

Key reference

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A list of affiliations appears at the end of the paper. e-mail: Adam_Rosenthal@med.unc.edu

Introduction

Development of the protocol

Transcriptome analysis of bacterial cultures is routinely used to study the variable expression of whole sets of genes and to identify the genes and regulatory features that allow bacteria to survive in variable environments and rapidly adapt to changing conditions. Numerous methods have been developed to capture the complete set of transcribed genetic information from a population of cells. However, these bulk measurements average the expression of multiple subpopulations of cells[1,2] and, thus, limit our understanding of heterogeneous behavior of different cell types present within a living system[3]. Recent breakthroughs in molecular biology (MB) techniques have overcome numerous technical challenges associated with analyzing the transcriptome at single-cell level and several common approaches exist to sequence the whole transcriptome of thousands of individual eukaryotic cells[4–14]. While these approaches work well on a variety of organisms, only a limited number of techniques are available for the analysis of single bacterial cells because of several inherent difficulties associated with transcriptomic measurements in microbes[15–20]. First, messenger RNA content in a bacterial cell is two orders of magnitude lower than that of a mammalian cell, ranging between 10^3 and 10^4 molecules[21]. Second, the mRNA half-life in bacteria is only a few minutes compared with several hours for mammalian mRNA[21]. Shorter mRNA half-life makes it difficult to capture a sufficiently large number of transcripts from individual cells and to analyze short-lasting molecular events. Third, unlike mammalian mRNA, bacteria do not exhibit extensive poly-adenylation of the 3′ end of transcripts ('poly-A tail'). While there are some cases of limited polyadenylation in bacteria[22], enrichment of total bacterial mRNA (<10% of cellular RNA) over the more abundant ribosomal RNA is thus not possible based on intrinsic poly-A tail tagging by hybridization to primers containing a 3′ poly-thymidine sequence[15,16,18,19,23]. Finally, single-cell techniques require cell lysis to capture transcripts, a difficult task in bacteria due to the presence of complex cell envelope architectures that may contain a cell wall, an outer membrane in Gram-negative bacteria and capsule in some species. To resolve these issues, bacterial single-cell RNA sequencing (scRNA-seq) techniques rely on cell fixation to preserve transcripts and methods for cell permeabilization with a combination of mild detergents and/or enzymatic degradation of the cell wall. RNA is captured through hybridization-based methods, which can rely on short (<10 nt) DNA primers of random sequences or longer (>50 nt) DNA probes with specific target sequences. Lastly, mRNA enrichment can be achieved by active depletion of ribosomal RNA[17,18], biased transcript modification[16] or obviated with the use of probes that directly capture the desired transcripts[19,23]. Each of these recently published techniques has advantages and drawbacks. The protocol outlined here, ProBac sequencing (ProBac-seq)[19], was devised to maximize capture of transcripts, avoid undesired products such as ribosomal RNA and provide end-user ease of use. This technique leverages the most commonly used and commercially available eukaryotic single-cell system (10x Genomics platform) to combine microfluidic droplet encapsulation of single bacterial cells with in situ hybridization of DNA probes for high sensitivity and throughput. The initial idea behind ProBac-seq relied on reports that single molecule florescence in situ hybridization (sm-FISH)[24] provides sensitive capture of quantitative data from fixed single bacterial cells[25], but the number of genes is typically limited by fluorescent channels and requires advanced imaging analysis. We reasoned that replacing the fluorophores with a sequence can provide a molecularly barcoded readout for thousands of probes in commercial droplet systems. Initial optimization experiments, not provided in this protocol, optimized the probe length to be similar to that used in commercial probe arrays (Agilent, for example), and the number of probes per gene was maximized to allow redundant measurements of the same target for increased specificity and potential removal of inaccurate probes and comparison of transcriptional intensity across gene and operon lengths. Probe penetration and hybridization temperature were empirically determined by testing different permutations. While hybridization temperature should work for probe sets designed with similar parameters against new species and strains, probe penetration conditions may need to be optimized for new organisms using FISH optimization workflows described

in literature[26–28] or referring to pre-existing FISH protocols for well-studied organisms. In our initial development, we selected rolling circle amplification (RCA) as a method to prepare large probe sets because this method inherently yields very high numbers of single-stranded DNA products without compounding errors as in traditional PCR[29].

Applications of the method

Bacterial scRNA-seq is emerging as a powerful tool for unbiased characterization of genome-wide transcriptomic differences from thousands of single cells within a population. These measurements can be applied to environmentally important bacterial species, industrial fermentation strains and a wide range of disease-causing species to reveal unknown cellular states. In the context of infectious disease, the sensitive transcript coverage of ProBac-seq can be used to decipher the responses of individual bacteria to environmental challenges, spatial gradients, host–immune responses and antibiotic exposure. In these conditions, several microbial populations may coexist in low abundance, making them difficult to detect using traditional bulk RNA sequencing. The capture of thousands of cells in a single assay allows rapid detection of rare cell types that can contribute to disease by overexpressing specific virulence factors (toxins, attachment and secretion systems) or pathways for evading antibiotics or immune detection (capsule formation, antigenic switching, various tolerance and persistence strategies)[16–20,23]. In the original ProBac-seq publication[19], we recover and validate distinct cellular states exhibited by fewer than 1% of the total cell population, such as sporulation by *Bacillus subtilis* in M9 media. In a larger cell cluster that is associated with genetic competence in *B. subtilis*, we identify 90% of the known competence regulon as differentially expressed within this cluster and additionally identify the regulatory structure of competence.

In addition to studies of infectious disease, there are several disciplines in bacteriology that are particularly well suited to employ ProBac-seq. Laboratory-based evolutionary experiments, which look at the emergence of genetic diversity in a population[30,31], can leverage the power of scRNA-seq to survey the extent of phenotypic heterogeneity in an evolved population and determine whether and how strategies that optimize phenotypic state switching emerge. These studies can further use evolutionary strategies to screen and select for regulatory features that control the amplitude and frequency of phenotypic switches.

Ecological studies using diverse bacterial species, including extremophiles, can identify common pathways and physiologies that are commonly segregated into specialized cells in response to specific environmental conditions. In the context of complex multispecies environments, such as the gut microbiome, single-cell studies can inform our understanding of the niche limits and range boundaries that cells from different species explore. In particular, these studies can determine whether and how these boundaries become more limited in the presence of particular signaling, environmental or organismal changes in the cells' surroundings. While ProBac-seq requires dedicated probe sets, which can be challenging to employ in multispecies mixtures, there are two approaches that allow for using probe sets in environmental contexts; first, the species of interest can be enriched after fixation using flow cytometry or other enrichment strategies, such as magnetic separation[32]. Enriched cells can then be probed with a strain-specific probe set to determine the niche plasticity of the target strain; second, probes for multiple interacting species can be used alongside probe markers for identifier strains (such as unique 16S rRNA sequences) in consortia of known species.

Overview of the procedure

To characterize distinct transcriptional states within a microbial population, we first determine unique 50 bp mRNA binding sequences with a recommended average of five sequences per transcript for each gene in the target organism's genome; although fewer or more probes can be designed to reduce cost or increase sensitivity of capture or percent sequence coverage of particular genes, as desired. With a typical bacterial genome, this equates to ~15,000–30,000 probes per strain. These binding regions are obtained either from previously designed microarrays or designed using existing bioinformatic tools[33]. Common DNA flanking sequences are added to the 5′ and 3′ ends of each mRNA-binding sequence that allow them to be circularized or for probe tagging and amplification during the single-cell

encapsulation procedure, and the resulting protoprobe sequences are then ordered from Twist Biosciences. To limit costs, unique flanking regions for ligation can be added to each library and several libraries (up to five) can be combined into one Twist order and selectively amplified for each library with the help of unique flanking region specific primers[34]. As a first step in probe amplification, the 5′-end of each protoprobe is phosphorylated with polynucleotide kinase to allow probe circular ligation, and the two ends of each protoprobe are ligated with T4 DNA ligase using a short oligo splint that brings the two ends together (Steps 1–3). The circular probe amplification templates are amplified by RCA with phi29 DNA polymerase and the resulting long products are digested with HindIII to produce linear protoprobe sets (Steps 4–7). The process is repeated twice more to prepare large probe quantities (Steps 8–24). In the final step of probe production, a unique molecular identifier (UMI) and poly-A tail are added to the 3′ end of the protoprobes by primer extension using a blocked extension template (Steps 25–26).

Once probes are prepared, the bacterial samples can be collected from culture media or environmental sources such as biofilms (Fig. 1). The cells are first fixed and permeabilized to allow probe penetration (Steps 27–34). The probes are then added to the bacterial suspension for hybridization to immobilized mRNA, and several washes are used to remove unhybridized or mis-hybridized probes (Steps 35–37). After these washes, the probed cells are counted using a flow cytometer to ensure there is a sufficient number for microfluidic encapsulation (Steps 38–41). The cells are then mixed with PCR reagents and primers in preparation for single-cell distribution into droplets (Steps 42–43). For the microfluidic encapsulation of cells with barcoded 10x gel beads, the probed cell suspension, gel beads and partitioning oil are loaded into the microfluidic gel beads in emulsion (GEMs) generation chip and run with the 10x chromium controller (Steps 44–46). After encapsulation, the GEMs are loaded into a thermocycler for in-droplet PCR labeling of probes with the droplet identification (ID) encoded in the 10x gel beads (Steps 47–52). Following the in-droplet labeling, the pooled single-cell probe library is processed with two subsequent PCR reactions that amplify the product and add Illumina adaptors and sample indices (Steps 53–58). The library size is then confirmed with an Agilent TapeStation which also provides the library concentration (Steps 59–60). The indexed libraries from different samples can be pooled to maintain the same molar ratio and then sequenced. Initial analysis using the 10x Cell Ranger software offers experimental metrics, population clusters and heat map of the differential gene expression (Fig. 1).

Comparison with other methods

Microscopic investigation of gene expression has used fluorescent reporters to identify bacterial cell types that differentially express genes for fundamental cellular processes including metabolism[35–37], cell adhesion and motility[38–43], virulence factors[42–46] and tolerance of antibiotics, among others. The advantage of using scRNA-seq to study heterogeneity in bacterial communities is that it provides an unbiased approach for investigating genome-wide expression levels in thousands of single cells, allowing for comprehensive analysis of gene–gene and cell–cell correlations. In the past 3 years, several bacterial scRNA-seq techniques, each with their own advantages and limitations, have been reported[14–19,23]. In MATQ sequencing, bacterial single cells are sorted into individual wells in multiwell plates and indexed to prepare sequencing libraries, providing high gene capture but an approach that is limited in terms of scale[13]. Another recently developed method, parallel sequential fluorescence in situ hybridization (par-SeqFISH), uses fluorescent microscopy to detect mRNAs through short, labeled secondary probes that can profile hundreds of thousands of bacteria in biofilms and provides spatial information about the location of each cell, but is limited to a number of genes detected at a time and requires highly specialized equipment and analysis methods[23]. One of the major advantages of ProBac-seq is that it provides a high capture of transcripts (several hundreds of transcripts per cell), even when bacteria are cultivated in minimal culture media, a condition under which there are normally a lower number of transcripts in each cell. Other reported methods have been benchmarked using conditions in which cells are grown in rich media[16,18] and most recover fewer transcripts per cell than ProBac-seq even in these growth conditions, indicating lower capture rates. Other methods such as prokaryotic expression profiling by tagging RNA in situ and sequencing (PETRI-seq)[15] and microbial split-pool ligation

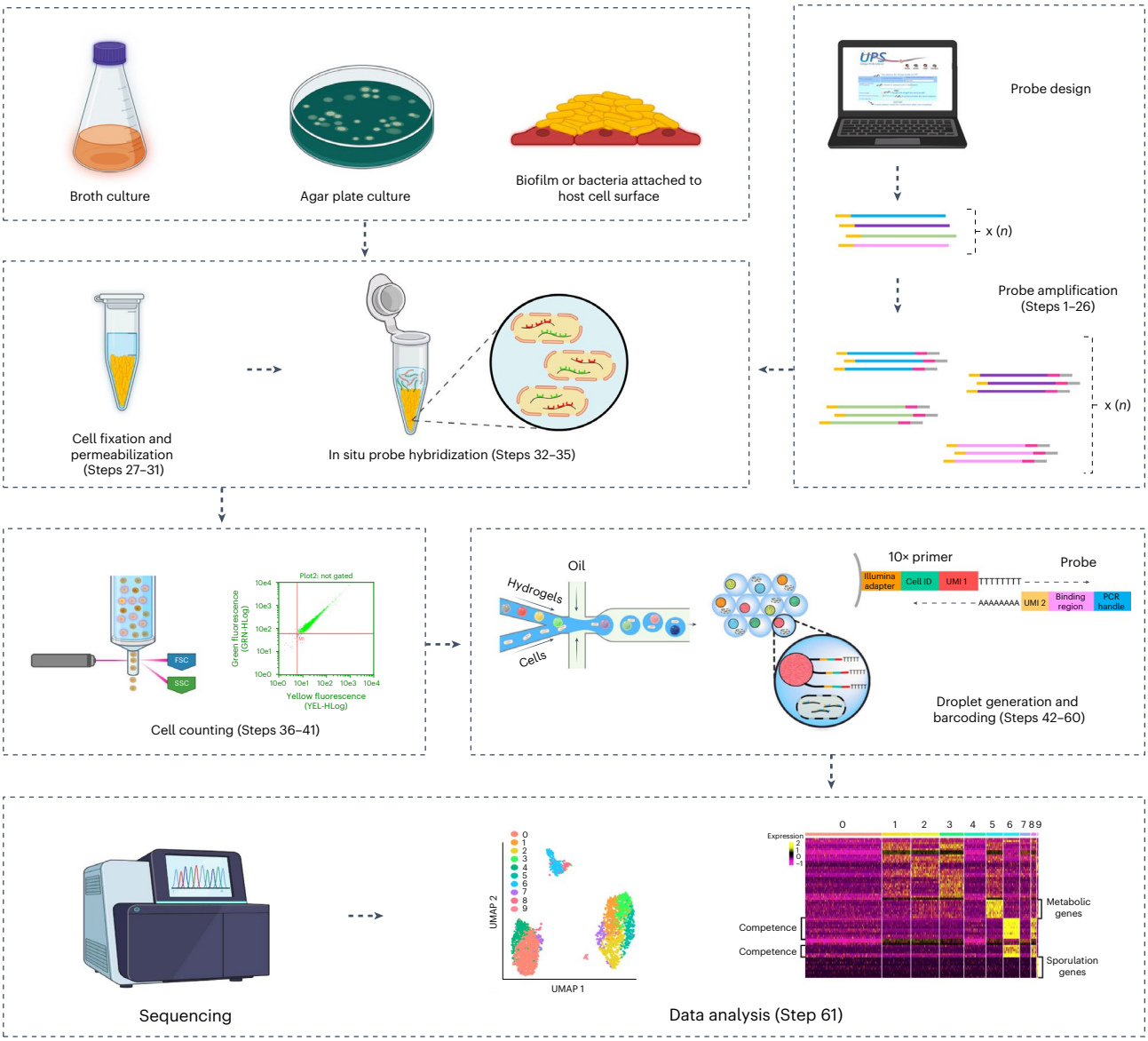


Fig. 1 | An overview of the ProBac-seq protocol. This protocol includes analysis of bacterial single-cell transcriptomes grown in liquid cultures, on agar plates and cells in biofilms attached to host cell surface. The genome-wide probe sets are designed with a web-based probe designing tool, ordered and amplified to achieve an ample supply of probes and ligated with UMIs and poly-A tails (Steps 1–26). The bacteria are immediately fixed to preserve their physiological state and permeabilized (Steps 27–31). The permeabilized bacteria are mixed with the probe pool for in situ hybridization (Steps 32–35). The bacteria with hybridized probes are enumerated using a flow cytometer and a green fluorescent dye (Steps 36–41). Encapsulation and unique barcoding of single cells are achieved with the 10x Genomics microfluidic device followed by probe library generation, sequencing and data analysis (Steps 42–61). The final barcoded probe libraries are sequenced using Illumina Nextgen sequencer, and the final data can be visualized as UMAP of heterogeneous cell populations and heat map of differentially expressed genes. The figure was created with BioRender.com. Panels with Steps 42–60 and Step 61 were reproduced with permission from ref. 19, Springer Nature Limited.

transcriptomics (micro-SPLiT)[16] can capture a fair number of transcripts (a median of 200–300 mRNA transcripts per cell) using split-pool combinatorial indexing technique but have thus far been limited to profiling bacteria grown in rich media, which consists of high number of transcriptionally active cells. In natural conditions, such as infection models, bacterial growth is much more similar to the slow growth observed in minimal media and, therefore, techniques that capture a low percentage of available transcripts are not sufficiently powered to investigate these samples, whereas ProBac-seq can be applied to diverse conditions including biofilms, infections and industrial settings[47].

Another advantage of ProBac-seq over alternative bacterial scRNA-seq protocols is that it provides a more accurate quantification of transcripts within a cell. In ProBac-seq, a UMI sequence allows for collapsing of sequence reads to count the true number of probes that hybridized to a predetermined 50 nt stretch of sequence within target transcripts, resulting in a quantitative measurement of transcript abundances within each single cell. While several other techniques include UMIs in their readout, these values serve as a proxy for transcript abundance[15–18,20] due to the use of short (<10 nt) random primers (randomers) and a reverse transcriptase to generate barcoded cDNA from primed transcripts. With this approach, multiple randomers can prime the same transcript at a given instance and, when combined with the strand displacement activity inherent to reverse transcriptase enzymes, will generate multiple copies of cDNA with different UMIs from the same transcript, leading to an inflation of true RNA counts and biased estimates of gene expression in downstream analyses. In ProBac-seq, because multiple copies of the same probe cannot occupy the same position on any given transcript and because sequencing libraries are made directly from the hybridized DNA probes without generating cDNA, the number of normalized UMIs are directly correlated with the number of mRNA targets present in each cell. Furthermore, the design of multiple probes targeting distinct regions per transcript allows for a higher chance of detecting any given mRNA molecule and provides additional independent measurements for each transcript that can be used to gain sufficient statistical significance for genes expressed at a low level.

Our probe libraries are generally ordered at subfemtomole quantities and amplified by RCA, which permits reamplifying probe sets for unlimited subsequent experiments resulting in low costs. This technique allows users to perform PCR inside 10x droplets using large probe sets, with a protocol that can be adapted to nonbacterial cells and applied to previously intractable systems such as cell types with unstable mRNAs, formalin-fixed and paraffin-embedded samples, intracellular parasites or endosymbionts, difficult to lyse eukaryotes and Archaea.

Limitations

Although ProBac-seq has advantages over other bacterial scRNA-seq techniques, the technique's reliance on the 10x platform and large oligonucleotide probe sets presents several limitations. Beside access to a 10x chromium controller and reagents, the most obvious limitation of this technique is that it requires prior knowledge of the genome and an upfront cost to design and order a large set of probes. ProBac-seq relies on probe hybridization, and hybridization across a target region (for instance, a gene) may not be entirely uniform. For instance, some regions may have increased secondary structure or occupancy potentially resulting in different probe binding affinities[48]. Additionally, probe misbinding or poor washes can potentially cause nontarget noise. To circumvent this, ProBac-seq uses multiple probes per transcript, allowing for the use of robust statistics (such as median signal) or the computational removal of nonrepresentative probes to minimize the effects of misbinding or poor binding probes. In addition to this computational approach, the HCR (hybridization chain reaction) wash buffer contains stringent off-binding additives, such as heparin (designed to discriminate against partial binding)[49], Denhardt's solution (which reduces nonspecific binding)[50], formamide (which is postulated to increase accessibility of probes to targets and reduce noise[51]) and other buffer components. Combined with stringent washes, we find a strong correlation ($r = 0.77$) between the signal from probe binding and signals from established, population-level bacterial RNA sequencing methods, which is in the same range as comparisons between microarrays and RNA sequencing[19]. However, there is also information that is missed in a probe-based approach, including the presence of nucleic acid mutations that may be present in transcripts such as single-nucleotide polymorphisms. The probe design parameters also mean that genes must be sufficiently unique and large enough to accommodate 50 bp regions with sufficient unique features. In our recent designs using software such as Unique Probe Selector (UPS) 2.0 (ref. 33) we are able to design probes for >95% of all annotated coding sequences. It is important to also keep in mind that older annotations of genomes may miss some newly identified genes or noncoding RNAs of interest, and no information on these genes will be discovered without a priori design of appropriate target probes. The probes also require amplification using a set of enzymatic reactions, which accrues an additional cost but

can provide a large supply of probes for many future experiments. Additionally, the reliance on probes means that a few short or highly homologous genes may not be targeted, and analysis is limited to known transcripts and therefore unannotated genes will be missed. Despite these limitations, ProBac-seq is fast, cost effective and yields high-resolution datasets with accurate transcript quantification.

Experimental design

Probe design and amplification (Steps 1–26)

ProBac-seq relies on the design of unique genome-wide probe sets composed of probes that share similar hybridization properties but that bind specifically to distinct target regions. A properly annotated whole-genome FASTA file for the organism of interest is used to check for sequence uniqueness, and a separate FASTA of desired targets (coding sequences or noncoding RNA transcripts) is used as the template to which probes that hybridize with the corresponding mRNA by sequence complementarity are designed. We previously used the online probe design tool UPS 2.0 (ref. 33) to select a predefined number of unique 50 bp regions from the coding regions using the default settings of a GC content between 40% and 60% and a melting temperature of 50–55 °C. This typically yields a set of ~14,000–30,000 probes per organism representing coding gene sequences ranging between 3,000 and 4,000 profiled genes depending on the type and genome size of the organism (Supplementary Table 1). In probe design software, such as UPS 2.0 or alternatives including eArray (Agilent), OligoMiner[52] or OligoArray[53], it is important to remove probes with overlapping sequences and probes that can bind to multiple positions on the genome. These can be filtered out in the output file provided by the program. Alternatively, for many model species, binding sequences can be obtained from previously published microarrays designed on the Agilent or similar platforms or designed using a number of other array design software tools[52,54–56]. These commercial designs provide validated probes ranging from 50 to ~65 bp, which can be used in ProBac-seq with only minor modifications that add the amplification and droplet-indexing probe regions. When designing probe sequences using this protocol, it is critical to make sure that the probe sequences, to which handle and extender sequence are appended, are in the correct orientation that will produce a final probe library that has a sequence complementary to the target mRNA. As our amplification scheme utilizes three rounds of RCA and each cycle reverses the direction of probes, we order primary probes in the same orientation as the gene coding sequences, which, after final RCA, become complementary to the coding region and bind with the transcripts. As an alternative to this computational strategy of designing and ordering probes as the coding sequences, an even number of rolling circle reactions can also be used to prepare final probes if we order reverse complement of the coding sequences.

For probe amplification and library preparation, two common sequences are added to mRNA binding sequences. A common 23 bp PCR handle sequence is added to the 5′ end of all probes and is used for circularization of the template during probe amplification and as a handle onto which PCR primers for library preparation attach during the single-cell encapsulation. Additionally, a short 17 bp extension handle sequence (extender) with a HindIII restriction site (AAGCTT) is added to the 3′-end. This 3′ sequence mediates circularization and relinearization processes during RCA and allows the addition of a UMI and poly-A tail to the 3′-end of each probe (Fig. 2). The use of HindIII in the amplification protocol can potentially interfere with probes that contain the AAGCTT sequence along the binding region. While we normally find that this only affects a handful of probes in a designed probe set, it is advisable to search for and remove all probes that contain this particular sequence in areas outside of the designed amplification loop.

Once designed, the probe libraries are ordered in subfemtomole quantities with a cost of ~US$0.10 per probe. This probe quantity is insufficient for even a single experimental sample; thus, amplification of the probe library by phi29 DNA polymerase-directed RCA is required. phi29 possesses high fidelity, extreme processivity and high-displacement ability. The first round of RCA (RCA-1) produces the complementary probe strand that can anneal to the mRNA transcript, but the yield is typically not sufficient for multiple experiments. Therefore, another two rounds of RCA (RCA-2 and RCA-3) are typically carried out to yield 0.35 pmol of each probe in the proper orientation to anneal to the mRNAs. The RCA-3 product undergoes

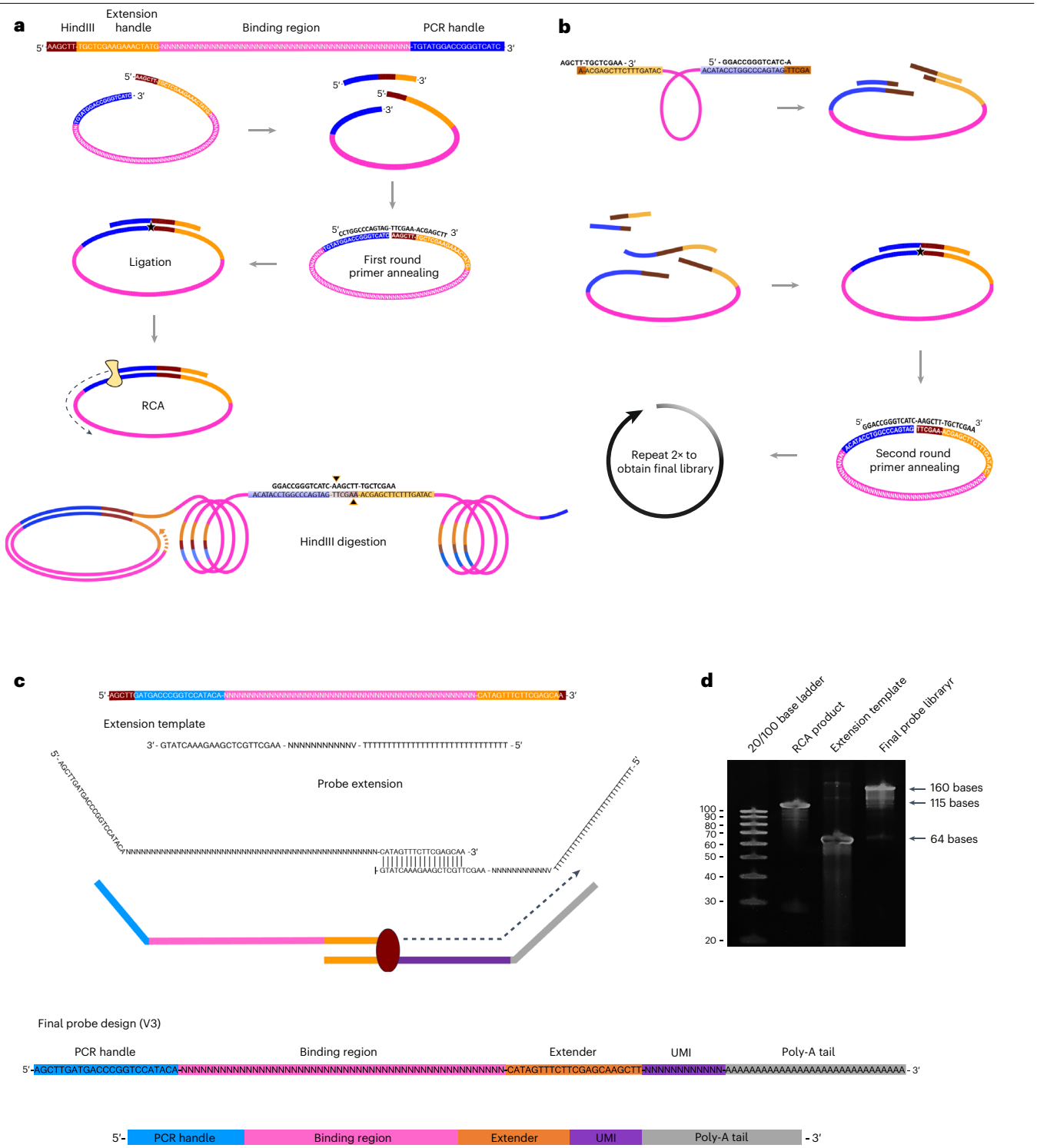


Fig. 2 | Preparation of oligonucleotide probe library. a, A commercially manufactured oligonucleotide pool of probes is circularized by a short oligonucleotide splint containing complementarity with the probe ends (5′ and 3′) and the free ends of the probe are ligated with T4 DNA ligase. The same splint primer is used in a RCA of the ligated probes using phi29 DNA polymerase followed by HindIII digestion to cut the probes into unit length single probe fragments with the help of a nicking primer. **b**, After cutting into single-probe lengths the probes are again circularized with the nicking primer splint and the ligation, RCA and HindIII digestion steps are repeated in two additional cycles to yield final library of protoprobes. **c**, A stretch of 12 random UMIs along with 30 T (thymidine) residues are added to the protoprobes with the help of an extension template primer with a blocked 3′ end to synthesize the final probe library with UMIs and poly-A tail. **d**, A 15% TBE–urea polyacrylamide gel representation of protoprobe, extension template and final probe library with a 20/100 base molecular size marker. The figure was created with BioRender.com.

HindIII digestion directed by a primer, which provides a double-stranded region needed for restriction enzyme digestion (Fig. 2). After digestion, the single-stranded protoprobes are purified using a 15% (wt/vol) TBE–urea gel. A multiple banding pattern is expected along with the desired probes because of the presence of other short oligos produced in the restriction enzyme digestion (Fig. 2). The desired banding region, ~100–120 bp, is excised from the gel with a razor blade and the fragment is crushed in a conical tube using a 1 ml serological pipette or a glass rod with a similar thickness. Elution buffer (EB) is added and incubated overnight at 37 °C with shaking to promote probe diffusion from the gel into the EB. After incubation, the solid gel particles are removed by a brief centrifugation, leaving the DNA probes in the supernatant. This supernatant is then concentrated by ethanol precipitation to recover an amplified protoprobe library with a concentration ranging from 500 to 1,000 ng/µl. Since bacterial mRNAs do not possess poly-A tails, which facilitate sequencing with the 10x Genomics kits, a poly-A stretch of 30 bases is added to each probe using an isothermal amplification (primer extension) reaction. A probe-specific UMI with random sequence of 12 bases is also added to the probes in this reaction and is positioned between the coding region and the poly-A stretch. This UMI helps to differentiate between copies of bound probes from final sequencing reads and reveals the true number of captured transcripts within each cell. Addition of the poly-A tail and UMI is performed by isothermal amplification with a 3′ blocked primer (with a phosphate moiety on the 3′-end), which contains a 5′ sequence homologous to the extension handle, 12 'N' bases encoding the UMI and a stretch of 30-thymine bases that are used to encode the poly-A ends in the resulting primer extension reaction (Fig. 2). This generates a large set of probes that contain different UMIs to aid counting of transcripts and a short unextended sequence of the blocked primer that can be separated from the probes using 15% (wt/vol) TBE–urea gel electrophoresis purification. After purification, the final product is concentrated by ethanol precipitation and the resulting probe solution can be stored at −20 °C long term.

Cell fixation and permeabilization (Steps 27–32)

A wide variety of Gram-positive and Gram-negative bacteria with regular cell walls are compatible with this method. We have profiled strains grown in liquid medium either aerobically or anaerobically with numerous testing conditions. For the purpose of this protocol, we utilize the Gram-positive *B. subtilis* as an example to test the in-population heterogeneity because it produces multiple known phenotypic subpopulations when grown in minimal media.

Since bacterial mRNAs have a short half-life, the cells must be chemically fixed before single-cell analysis to inhibit cellular activity, preserve transcripts and reinforce structural integrity, while allowing the movement of small molecules such as probes (Fig. 1). Our method uses 1% paraformaldehyde for immediate fixation to retain cellular states before downstream manipulation[57–59]. Paraformaldehyde works by creating covalent bonds between proteins, thus immobilizing them in situ while conserving their spatial structure and arresting cellular processes. A few scRNA-seq methods use heat to reverse formaldehyde crosslinking[58,59]. However, reverse crosslinking is often inefficient and can result in high loss of mRNA molecules and RNA shearing[59]. Since our method relies on a poly-dA sequence included on each probe, we do not require an additional reverse fixation step. Fixed cells can be stored in methanol:acetic acid (4:1 (vol/vol), MAAM) solution for up to 1 week at −20 °C.

Gram-positive bacteria possess a thick layer of peptidoglycan in the cell wall compared with Gram-negative bacteria, and hence, the permeabilization step may vary for some species. While a gentle lysozyme treatment is enough for efficient permeabilization of many Gram-negative bacteria we tested, an optimized treatment with additional cell wall-degrading enzymes, including lysostaphin and mutanolysin, may need to be established for effective analysis of Gram-positive bacteria. After cell wall digestion, the samples are washed with PBS with 0.1% Tween to inactivate lysis enzymes and to further permeabilize cell membranes.

Probe hybridization (Steps 33–35)

Permeabilized cells are suspended in probe hybridization buffer and incubated at 50 °C as a prehybridization step. The amount of bacterial pellet is monitored carefully to ensure there is a sufficient amount (Fig. 3a). Hybridization of probes is achieved at 50 °C to match the optimal

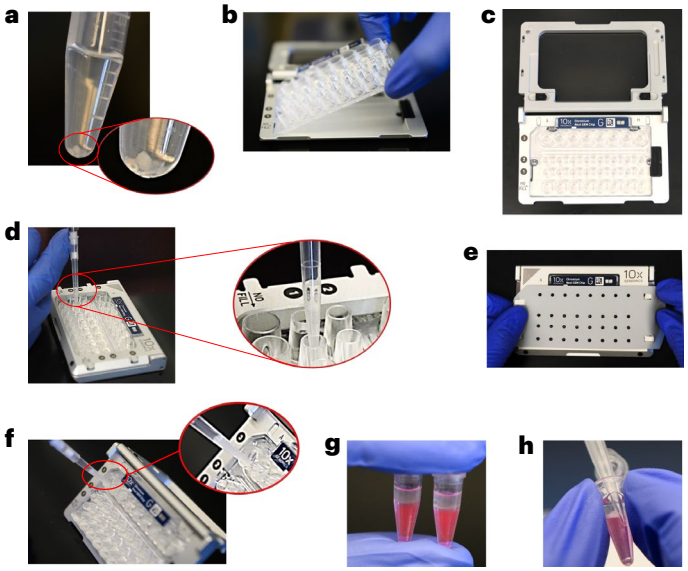


Fig. 3 | Crucial steps in ProBac-seq. a, A representation of the approximate sample pellet size after probe hybridization assuming the cell size as the size of most of the Gram-positive and Gram-negative bacteria, which is ~1–2 µm (Step 36). **b**, Assembly of the 10x microfluidic chip in the chip holder with first placing of left-hand side of the chip (Step 44). **c**, Proper placement chip within a chip holder (Step 44). **d**, Loading of sample and master mix to the chip well (Step 45). **e**, Placement of gasket aligning the upper-left corner notch on the holder (Step 45). **f**, Slow and careful aspiration of GEMs after encapsulation of single cells into droplets (Step 47). **g**, Biphasic layer after in-droplet PCR amplification and addition of recovery reagent (pink) (Step 50). **h**, Removal of bottom-pink layer with a 200 µl pipette inserting the tip through the upper aqueous layer (Step 51).

annealing temperature designed in the probe selection process. It is also important to heat the final probe library briefly at 96 °C and then cool to room temperature (24–26 °C) to dissociate probes from each other. The preincubated probes and cells in hybridization buffer are then mixed and gently pipetted twice to distribute probes. The probe hybridization buffer contains dextran sulfate and formamide, which help in accelerating the hybridization process by locally increasing the concentration of interacting molecules without increasing background signal from nonspecific binding[60]. To remove unbound and misbound probes, several washes are performed at 50 °C with probe-wash solution that lacks dextran sulfate. Components of the probe hybridization and wash solutions may interfere with the downstream enzymatic reactions or the encapsulation process on the 10x device resulting in a clog and improper droplet generation. Therefore, hybridized cells are washed with PBS three times to ensure the removal of interfering buffer components and finally resuspended in PBS.

Cell counting (Steps 36–41)

A number of manual and automated methods are available for counting cells, including microscopically through use of a hemocytometer or similar microscopic counting chambers, automated cell counters and/or flow cytometry. Here, we use a Guava easyCyte HT flow cytometer for efficient counting using SYTO-9, a green fluorescent stain that binds nonspecifically to nucleic acids and has an emission spectrum of ~500 nm. We optimized the dilution of SYTO-9 to 1:600 in PBS for efficient visualization of cells with green fluorescence. PBS with SYTO-9 has been used as a blank for background noise cancellation. Forward scatter height (FSC-H) and side scatter height (SSC-H) measurements correlate with the relative size and granularity/complexity of detected cells, respectively. Depending on the bacterial cell size and shape, as well as the instrument, forward scatter (FSC) and side scatter (SSC) gains must be adjusted to capture cell events within instrument bounds and confidently differentiate between cells and debris. Typically, using the Guava easyCyte HT, FSC and SSC gains of ~800 have been used for the detection of rod-shaped bacilli (Supplementary Fig. 1). Also, a typical threshold for selecting the bacterial population has been set to a fluorescein isothiocyanate (FITC) gain of >500 for the SYTO-9-stained whole cells (Supplementary Fig. 1). This step is carefully monitored and optimized since continuing with an underestimation of the true number of cells in solution results in high number of multiplets, that is, multiple cells within a single droplet that produce signal that is not single-cell data. Conversely, overestimating the number of prepared cells results in sparse datasets, which make clustering and imputation of data for genes expressed to a low level difficult.

Microfluidic encapsulation and barcoding (Steps 42–60)

ProBac-seq uses the 10x Genomics chromium controller and chromium single cell 3′ reagents kit (V3 chemistry) for single-cell partitioning and barcoding with several modifications for bacterial cells (Fig. 3b–f). A PCR master mix is prepared to add with the cell suspension for the amplification of hybridized probe sequences within each droplet after cell encapsulation. This PCR mix replaces the standard 10x reagents to allow in-droplet barcoding and amplification of DNA based probes using an 18 bp 'InDrop' reverse primer complementary to the PCR handle on the 5′ end of each probe (Supplementary Table 2) and the 10x hydrogel-bound forward primer containing a 3′ poly-dT capture sequence and cell barcode (Supplementary Table 2). The emulsion formed during GEM generation produces an opaque homogeneous solution of ~100 µl, and care should be taken when it is drawn up with a pipette slowly over a course of 20 s and transferred to a PCR tube to slowly release the contents to avoid the possibility of GEM disruption (Fig. 3f). The tubes with a well-distributed emulsion are immediately placed on a thermal cycler for a brief (six-cycle) PCR amplification, during which the probes within a given droplet are tagged with a unique cell barcode. The PCR-1 cycle initiates with heating at 94 °C for 5 min, which melts the hydrogels and dissociates the 10x poly-dT primers into the droplet, enabling the reaction. The PCR annealing temperature is set at 50 °C due to the low melting temperature of poly-A priming.

After PCR-1, microfluidic emulsions are broken by adding the 10x recovery agent. The resulting biphasic mixture includes an upper aqueous phase containing the probes with other short nucleic acids and enzyme, whereas the pink bottom phase contains partitioning oil and recovery agent (Fig. 3g,h). A pipette is carefully inserted through the upper aqueous layer to remove the pink bottom layer leaving 10–15 µl of pink solution at the bottom, ensuring that no probe library product is lost. Formation of a cloudy upper aqueous layer does not mean an improper GEM generation and can be overcome with a brief vortex and a short spin for 10 s. An aqueous phase volume lower than 100 µl indicates a possible clog during the GEM generation step or problems with the microfluidic pressure control. Purification of the pooled DNA is performed with the NucleoSpin gel and PCR clean-up kit (Macherey-Nagel) instead of the 10x Dynabeads MyOne SILANE kit because the ProBac-seq product is double stranded, whereas traditional 10x libraries contain single-stranded cDNA from reverse transcription. Nucleospin purification is carried out using manufacturer's protocol for a 100 µl reaction volume.

Half of the purified product can be stored as backup at −20 °C for several months, and the other half is further amplified with the same InDrop reverse primer used in the encapsulation reaction and a PCR-2 forward primer that is complementary to the 5′ end of the 10x gel-bead primer but does not contain the 10x cell ID or UMI sequences and, thus, does not introduce any additional information to the library. This PCR-2 scale-up reaction is amplified for 16 cycles and is followed by a purification step using the NucleoSpin kit (Macherey-Nagel) to enrich and purify the desired library pool. A 1.2% (wt/vol) agarose gel run is performed to ensure the appropriate library amplification analyzing the size (~200 bp) and intensity of the band. A TapeStation is then used to quantify the DNA products and determine the initial template amount for the next amplification reaction.

Illumina next-generation sequencing requires specific 5′ and 3′ library ends to aid the sequencing process. Therefore, a final library preparation step adds Illumina adaptors using a low cycle number PCR-3 with forward and reverse primers containing the Illumina i5 and i7 adapters. For multiplexing of up to 16 samples, a unique 8 bp index sequence is added to the conventional reverse primer as indicated in Supplementary Table 2. These index primers mark all reads within a sample with the same index sequences to distinguish them from other pooled samples. After PCR-3, the product must be purified with the NucleoSpin clean-up kit (Macherey-Nagel) and analyzed on a TapeStation for final estimation of the probe library (Supplementary Fig. 2). Efficient library preparation should include a single band of ~250 bp and concentration of ~3–5 ng/µl to process with the Illumina sequencing platform. While pooling multiple samples with different indexes, we prepare an equimolar concentration of samples for libraries with similar cell numbers and expected coverage. This mixed library is again run on a TapeStation for final quantification and to confirm the band sizes. In cases where primer dimers or other undesired bands exist, the correct band can be further purified using magnetic DNA

purification beads (SPRIselect, Beckman Coulter) or agarose gel electrophoresis and extraction using the NucleoSpin clean-up kit (Macherey-Nagel) using standard published protocols.

The libraries can be sequenced on Illumina compatible platforms. We typically use the NextSeq-2000 instrument with 100 cycle reagents. Libraries are spiked with 30% PhiX and sequenced for 8 cycles in the i7 index direction and 119 cycles for 'read 1'. The addition of 30% PhiX improves fastQ quality scores.

Data analysis

The analysis of each bacterial species requires dedicated reference files that specify the genome sequence and feature (that is, genes or probes) annotations. These reference files are created with the 10x Genomics Cell Ranger pipeline using the make-reference (mkref) command and allow mapping and counting of UMI-corrected reads onto each unique probe design in the library. We have deposited Python and MATLAB code (https://gitlab.com/hormozlab) that transform a list of probes into the two files (.fasta and .gtf) required by the Cell Ranger mkref command.

In addition to creating the reference genome, the Cell Ranger package accepts paired-end reads that have one 28 bp read (read 1) corresponding to the cell ID and UMI and a second read (read 2), which is mapped onto the reference genome. The sequencing strategy we employ returns both sequences in one long read (read 1, 119 bp) and necessitates splitting this read into two sequences with the option to swap the 10x-based UMI with the UMI added onto each probe. This is the recommended way to analyze ProBac-seq data, since the use of multiple cycles in PCR-1 can result in several copies of 10x primer amplifying the same probe, and thus, using the standard 10x UMI will overcount transcripts. An accurate way to count transcripts in ProBac-seq is, therefore, to replace the 10x UMI sequences, thus relying on the probe-based UMI to collapse the reads. Scripts in both MATLAB and Python for this UMI-swapping and read-splitting can be found in the Hormoz lab gitlab page (https://gitlab.com/hormozlab).

Once the reference genome and UMI-swapped FASTQ files are created, libraries can be processed using the Cell Ranger count command. The resulting web summary file (.html) provides basic control metrics regarding sequence quality, read alignment, barcode detection, gene expression and cell clustering. Additionally, the .cloupe output files can be viewed on the Cell Ranger Loupe Browser to further explore cell clustering and differential gene expression. For more complex analysis using Python or R, the .h5 files in the Cell Ranger output folder can be imported into programs such as Scanpy and Seurat. Finally, the Cell Ranger algorithm for distinguishing true cell barcodes from background (cell calling) is designed for mammalian cells and may not perform optimally on bacterial data. To address this, we have developed more sophisticated algorithms for cell calling that can be found at the Hormoz lab Gitlab page (http://gitlab.com/hormozlab).

Materials

Biological materials

- Bacterial strains (*Bacillus subtilis* strain 168 ATCC cat. no. 23857) (grown and fixed in the desired biological conditions; for further information, see 'Cell fixation and permeabilization' in 'Experimental design')
  ▲ **CRITICAL** Samples should be fixed immediately to preserve transcription at a fixed timepoint.

Reagents

▲ **CAUTION** Refer to the safety information provided by the vendors and take necessary precautions when handling reagents and equipment.

- Oligonucleotide probes (Twist Biosciences, special order)
  ▲ **CRITICAL** Twist Biosciences makes the probes using high-fidelity synthesis, which offers uniform yield with low error rates.
- Primers (Integrated DNA Technologies, IDT, special order; Supplementary Table 2)
- T4 polynucleotide kinase (10 U/µl; New England Biolabs, NEB, cat. no. M0201)

- T4 DNA ligase reaction buffer (10×; NEB, cat. no. B0202)
- MB-grade water (Corning, cat. no. 46-000-CI)
- T4 DNA ligase (NEB, cat. no. M0202)
- phi29 DNA polymerase (Biosearch Technologies, cat. no. F83900-1)
- phi29 DNA polymerase reaction buffer (Biosearch Technologies, cat. no. 88901-1)
- Dithiothreitol (DTT; Thermo Fisher, cat. no. P-2325)
- dNTP mix 25mM (Thermo Scientific, cat. no. R1121)
- Bovine serum albumin (BSA; NEB, cat. no. B9200S)
- HindIII-HF (NEB, cat. no. R3104L)
- NEB reaction buffer 2 (NEB, cat. no. B7002S)
- DNA polymerase I, large (Klenow) fragment (NEB, cat. no. M0210L)
- 3 M sodium acetate (Invitrogen, cat. no. AM9740)
- Ethanol (Decon Labs, cat. no. 2716)
- Buffer EB (Qiagen, cat. no. 19086)
- Adenosine 5′-triphosphate (ATP; NEB, cat. no. PV3227)
- 16% (vol/vol) Paraformaldehyde (Electron Microscopy Sciences, cat. no. 15710)
- UltraPure SSC, 20× (Invitrogen, cat. no. 15557-044)
- Methanol (OmniSolv, cat. no. MX0488-6)
- Acetic acid (Thermo scientific, cat. no. 035566.K2)
- Phosphate buffered saline (PBS; Fisher BioReagents, cat. no. BP399-1)
- Tween 20 (Sigma-Aldrich, cat. no. P1379)
- UltraPure 10× TBE buffer (Invitrogen, cat. no. 15581-028)
- Oligo length standards 20/100 ladder (IDT, cat. no. 51-05-15-02)
- SYBR Gold nucleic acid stain (10,000×; Invitrogen, cat. no. S11494)
- Ready-Lyse lysozyme solution (Biosearch Technologies, cat. no. R1804M)
- (Optional) Mutanolysin from *Streptomyces globisporus* ATCC 21553 (Sigma, cat. no. M9901-10KU), for use with certain species if needed
- (Optional) Lysostaphin from *Staphylococcus staphylolyticus* (Sigma, cat. no. L9043-.5MG), for use with certain species such as *Staphylococcus aureus*
- HCR probe hybridization buffer (50% (vol/vol) formamide, 5× SSC, 9 mM citric acid (pH 6.0), 0.1% (vol/vol) Tween 20, 50 µg/ml heparin, 1× Denhardt's solution and 10% (wt/vol) low molecular weight dextran sulfate, in MB-grade water). Sample type: cells in suspension (Molecular Instruments, cat. no. BPH02624)
- HCR probe wash buffer (50% (vol/vol) formamide, 5× SSC, 9 mM citric acid (pH 6.0), 0.1% (vol/vol) Tween 20 and 50 µg/ml heparin, in MB-grade water) (Molecular Instruments, cat. no. BPW02125)
- SYTO-9 green fluorescent nucleic acid stain (Invitrogen, cat. no. S11494)
- Guava instrument cleaning fluid (ICF; Cytek Biosciences, cat. no. 42000140)
- Household bleach (Clorox, for flow cytometry decontamination)
- UltraPure glycerol (Invitrogen, cat. no. 15514011)
- Q5 Hot Start High-Fidelity 2× Master Mix (NEB, cat. no. M0494S)
- Agilent D5000 TapeStation reagents
  - D5000 ladder (Agilent, cat. no. 5190-7746)
  - D5000 sample buffer (Agilent, cat. no. 5190-7744)
  - D5000 ScreenTape (Agilent, cat. no. 5067-5588)
- ddPCR multiplex supermix, 4× (BioRad, cat. no. 12005909)
  ▲ **CRITICAL** The BioRad ddPCR mix is formulated to allow PCR in droplets and has proprietary reagents that help with droplet stability to thermal cycling. Other ddPCR mixes with the same property may be substituted.
- V3′ Next GEM microfluidic chip G kit (10x Genomics, cat. no. PN-1000128)
  ▲ **CRITICAL** The protocol relies on the carefully calibrated microfluidic encapsulation conditions provided by the commercial 10x Genomics reagents.
- 10x V3.1 single cell reagent kit (10x Genomics, cat. no. 2000164)
  ▲ **CRITICAL** ProBac-seq droplet labeling relies on the 10x Genomics molecular barcoding scheme that uses the 10x 3′ labeling reagents.
- NucleoSpin Gel and PCR clean-up kit (Macherey-Nagel, cat. no. 740609.250)

- Oligo-binding buffer NTC (Macherey-Nagel, cat. no. 740654.100)
- Gene Ruler 50 bp DNA ladder (Thermo Scientific, cat. no. SM0371)
- Gel loading dye purple (6×; NEB, cat. no. B7024A)
- NextSeq 1000/2000 P2 reagents (Illumina, cat. no. 20046881)
- 15% Invitrogen Novex TBE–urea gel (Invitrogen, cat. no. EC6885BOX)
- Novex TBE–urea sample buffer (2×; Thermo Fisher Scientific LC6678)
- SPRIselect (Beckman Coulter, cat. no. B23318)
- AMPure XP (Beckman Coulter, cat. no. A63880)

Equipment

- Conical centrifuge tubes, 50 ml and 15 ml; (Falcon, cat. nos. 15352070 and 325096)
- Culture-treated microplate, 96-well (Falcon, cat. no. 3367)
- Disposable scalpel (Fisher scientific, cat. no. 22-079-707)
- Nitrile gloves, S–XL (Fisher scientific, cat. no. 19-130-1597)
- Microcentrifuge tube, 1.5 ml (Fisher scientific, cat. no. 05-408-129)
- Square Petri dish with lid, 10 cm (Electron Microscopy Sciences, cat. no. 70691)
- Pipette tips, 10 µl, 20 µl, 300 µl and 1,000 µl (Genessee Scientific, cat. nos. 26-401, 24-401, 23-417 and 24-430, respectively)
- Razor blade (VWR, cat. no. 55411-050)
- Round-bottom polystyrene test tubes with 35 µm cell strainer snap cap (Falcon, cat. no. 352235)
- Centrifuge (Eppendorf Centrifuge 5425, cat. no. 5405000646)
- Microcentrifuge (Prima mini microcentrifuge, model MIDSCI, cat. no. MF8-C)
- Incubator shaker (Benchmark Scientific, cat. no. H2010)
- TapeStation bioanalyzer (Agilent, cat. no. G2991BA)
- Flow cytometer (Millipore, Guava easyCyte HT, cat. no. 0500-4005)
- (Optional) Petroff–Hausser counting chamber (Hausser Scientific, cat. no. 3900)
- Pipettor, 10 µl, 20 µl, 200 µl and 1,000 µl (Genesee Scientific, cat. nos. 33-GSC-P10T, 33-GSC-P20, 33-GSC-P200T and 33-GSC-P1KT, respectively)
- NanoDrop OneC spectrophotometer (Thermo Scientific; cat. no. 840274200)
- Vortex mixer (Fisher Scientific, cat. no. 12-812)
- Thermal cycler (Eppendorf, cat. no. 6313000018)
- Heat block (Fisher Scientific, cat. no. 88-860-021)
- Refrigerators, 4, −20 and −80 °C (Fisher Scientific, cat. nos. FBG505GA, 05LFEEFSA and RDE30086FA, respectively)
- 10x Chromium controller (Series X; 10x Genomics, cat. no. CXVG10754)
- XCell SureLock Electrophoresis Cell (Thermo Scientific, cat. no. EI0001)
- PowerEase power pack (Invitrogen, cat. no. PSC350SB)
- Lonza FlashGel DNA starter kit (Lonza, cat. no. 57026)
- Transilluminator (TruBlu 2 blue/white transilluminator, EDVOTEK, cat. no. 557)
- Blue light-blocking orange safety goggles (EDVOTEK, cat. no. 631- O)
- Kimwipes (Fisher Scientific, cat. no. 06-666)
- PCR tube strip (USA Scientific, cat. no. 1402-4700)
- Thermomixer (Eppendorf, cat. no. 5382000023)
- Supreme air fume hood (Kewaunee)
- Class II type A2 biological safety cabinet (Fisher scientific, cat. no. 13-261-223)

Reagent setup

Probe and primer reconstitution

Dissolve probes and primers in an appropriate amount of MB-grade water as per the supplier's instructions to make stock concentrations of 100 µM or 1 mM as applicable. Can be stored at −20 °C at least for a year.

MAAM

Mix four volumes of methanol and one volume of acetic acid to prepare the MAAM solution. The MAAM solution is stable at room temperature for several months.

70% (vol/vol) and 75% (vol/vol) ethanol
Prepare ethanol solutions freshly by diluting absolute ethanol in MB-grade water for every use and avoid storing.

Wash buffer (Macherey-Nagel kit)
Add an appropriate amount of absolute ethanol to the wash buffer mentioned in the kit and store at room temperature for at least a year in a screw-capped bottle.

Gel staining solution
Add 5 µl of SYBR Gold nucleic acid stain (10,000×) in 50 ml of 1× TBE buffer to stain the 15% (wt/vol) TBE–urea gel. Since the dye is light sensitive, prepare the stain freshly at room temperature before staining and stain the gel in the dark.

Equipment setup

15% (wt/vol) TBE–urea gel assembly
Remove the gel from the sealed packet and carefully detach the well partition comb. Break the comb to one-third of its original size and use as a cut comb. With the help of the cut comb or small metal spatula, carefully break and remove the well partitions by inserting them into the gel wells and sliding one corner to another to convert a ten-well gel to a single broad well. Place the gel in the running chamber keeping the top open side facing inward to allow contact with the running buffer. Place the blocking plate on the opposite side of the gel and lock the gel tension wedge. Fill the chamber with 1× TBE buffer and load the sample to run at 180 V for 50 min.

1.2% (wt/vol) Agarose gel system (Lonza Flash Gel)
Remove the well seal from the gel and set on the dock. Load the samples and run at a constant voltage of 200 V for 20 min. Visualize bands directly by pressing the lamp button on the dock.

10x Chromium next GEM chip G assembly
Assemble the 10x GEM chip according to the manufacturer's specifications. Briefly, remove the chip from the sealed pack and place the left-hand side of it to the holder aligning the top-left corner notch. Gently press the right-hand side of the chip until hearing a click sound indicating that the chip is locked properly (Fig. 3b,c). Load the samples according to the 10x Genomics instructions.

Flow cytometer
Before running the samples, run two cleaning cycles following the manufacturer's instructions to ensure the flow cell is completely clean. Load samples in addition to one blank tube with 1× PBS and SYTO-9 dye to determine background noise in analysis. Run the samples at the preset 'low flow rate' setting on the Guava software, with events and time limits of 5,000–10,000 events and 200 s, set the Guava sample mixing setting for mixing 3 s at a high revolutions per minute (rpm) preset.

Procedure

Probe preparation

Phosphorylation of the oligonucleotide pool

● **TIMING 1 h**

1. Phosphorylation of the oligonucleotide library is a necessary first step. To begin probe amplification, dilute the Twist oligo library containing ~30,000 probes in MB-grade water to produce a 0.5 µM starting stock. Prepare a 10 µl reaction in a PCR tube as follows for the probe library.

Component	Amount (µl)	Final concentration
Crude oligonucleotide library	2.0	0.1 µM
10× T4 DNA ligase reaction buffer	1.0	1×
T4 polynucleotide kinase (10 U/µl)	0.2	0.2 U/µl
MB-grade water	6.8	
Total	10.0	

Incubate in a thermal cycler at 37 °C for 30 min, followed by heat inactivation at 65 °C for 20 min. Keep the samples on ice until proceeding to the next step.
▲ **CRITICAL STEP** When working with enzymes, it is important to keep them on ice. Other reagents should be thawed slowly on ice. Wear gloves while setting up the reaction.
■ **PAUSE POINT** The phosphorylated reaction can be frozen at −20 °C indefinitely for future amplification once inactivated

First-round probe amplification

● **TIMING 17 h**

2. Annealing of first-round RCA (RCA-1) primer. Prepare a master mix in a PCR tube to add the following per reaction to obtain a total volume of 100 µl per sample:

Component	Amount (µl)	Final concentration
Phosphorylated oligonucleotide pool	2.0	
10× T4 DNA ligase reaction buffer	10.0	1×
MB-grade water	87.0	
First-round RCA primer (0.3 µM)	1.0	3.0 nM
Total	100.0	

Incubate at 95 °C for 2 min and then cool to 10 °C for 5 min in a thermal cycler.
▲ **CRITICAL STEP** the primer concentration used in each round of amplification is different due to different template amounts present, take care to use the recommended concentration.

3. Annealing of first-round RCA primer is followed by a ligation step to join both the ends of the probes creating a circular template for amplification. For ligation, add the following to the 100 µl reaction from Step 2 to make a final volume of 200 µl for each sample:

Component	Amount (µl)	Final concentration
First-round primer annealed library pool (from Step 2)	100.0	
MB-grade water	79.0	
10× T4 DNA ligase reaction buffer	10.0	
ATP (10 mM)	10.0	0.5 mM
T4 DNA ligase (120 U/µl)	1.0	0.6 U/µl
Total	200.0	

Incubate at room temperature for 20 min, followed by heat inactivation at 65 °C for 10 min.

4. First RCA (RCA-1). Amplify the ligated probes using phi29 DNA polymerase, which has 5′–3′ polymerase and strand displacement activities. Prepare a master mix to add 400 µl to the 200 µl of ligated products from Step 3 for each of the samples.

Component	Amount (µl)	Final concentration
Ligated product from Step 3	200.0	
MB grade water	329.0	
10× phi29 buffer	40.0	1×[a]

Component	Amount (µl)	Final concentration
100× rBSA	6.0	1×
dNTP mix (25 mM each)	17.0	0.7 mM each
phi29 DNA polymerase (10 U/µl)	8.0	0.13 U/µl
Total	600.0	

[a]The final concentration of the buffer is calculated based on the total reaction volume excluding the volume of the product used from the previous step since the reaction from the previous step contains buffer with a similar composition.

Incubate the total 600 µl reactions at 30 °C for 14 h, and heat inactivate at 65 °C for 10 min. Keep the samples at 4 °C until ready for the next reaction.

■ **PAUSE POINT** The inactivated sample can be stored overnight at 4 °C.

5. Annealing of second-round RCA (RCA-2) primer. Add 10 µl of 100 µM second-round RCA primer to achieve a total volume of 610 µl reaction per probe set. Incubate at 95 °C for 2 min on a heat block, and then cool down to room temperature for 10 min.
6. Restriction digestion with HindIII: to digest with HindIII, prepare a master mix for each sample in a 15 ml conical tube as follows:

Component	Amount (µl)	Final concentration
MB-grade water	870.0	
NEBuffer 2 (10×)	100.0	1×
HindIII–HF (20 U/µl)	30.0	0.6 U/µl
Total	1,000.0	

Add the 610 µl reaction from Step 5 to the 1 ml of master mix and incubate at 37 °C for 1 h in an incubator, followed by heat inactivation at 65 °C for 20 min on a heat block. Cool down to room temperature for 15 min.

7. Product purification. Clean the digested products with Macherey-Nagel cleanup kit according to the manufacturer's protocol and elute in 50 µl of MB-grade water. Apply the optional oligo-binding buffer NTC for probe binding to the purification column instead of buffer NTI used for the purification of double-stranded DNA.

■ **PAUSE POINT** The purified product can be stored at −20 °C until the next day.

Second-round probe amplification

● **TIMING 18 h**

8. Ligation. Add 10 µl of 100 µM second-round RCA primer to achieve a total volume of 60 µl. Incubate at 95 °C for 2 min and then cool to 10 °C for 5 min in a thermal cycler for efficient annealing. Prepare a ligation reaction adding the following for each reaction:

Component	Amount (µl)	Final concentration
Second-round primer annealed probe pool	60.0	
10× T4 DNA ligase reaction buffer	7.0	1×
ATP (10 mM)	1.0	0.14 mM
T4 DNA ligase (120 U/µl)	2.0	3.4 U/µl
Total	70.0	

Incubate for 20 min at room temperature, heat inactivate for 10 min at 65 °C and cool to room temperature.

9. Second RCA (RCA-2). Set up a reaction mix to add 350 µl to the reaction from Step 8.

Component	Amount (µl)	Final concentration
Ligated product from Step 8	70.0	
MB-grade water	297.5	
10× phi29 buffer	35.0	1×[a]

Component	Amount (µl)	Final concentration
100× rBSA	3.5	1×[a]
dNTP mix (25 mM each)	10.0	0.7 mM each
Phi29 DNA polymerase (10 U/µl)	4.0	0.1 U/µl
Total	420.0	

[a]The final concentration of the reagents is based on 350 µl reaction volume.

Incubate for 14 h at 30 °C and heat inactivate for 10 min at 65 °C. Store the samples at 4 °C until the next reaction.

◆ **TROUBLESHOOTING**

10. Annealing of third-round RCA primer. Add 10 µl of 1 mM third-round RCA primer (which is the same primer used in the first-round annealing) to the reaction from Step 9 to get a total volume of 430 µl for each probe set. Incubate for 2 min at 95 °C and cool down to room temperature.
11. Restriction digestion with HindIII. For the HindIII digestion, prepare a master mix for each sample adding MB-grade water, NEBuffer 2, HindIII restriction enzyme and the reaction from Step 10 to achieve a total volume of 1.23 ml. Incubate for 1 h at 37 °C, followed by heat inactivation for 20 min at 65 °C and cooling to room temperature for 15 min.

Component	Amount (µl)	Final concentration
Third-round primer annealed suspension from Step 10	430.0	
MB-grade water	710.0	
NEBuffer 2 (10×)	80.0	1×[a]
HindIII–HF (20 U/µl)	10.0	0.16 U/µl
Total	1,230.0	

[a]The final buffer concentration is based on 800 µl of total reaction volume.

◆ **TROUBLESHOOTING**

12. Before proceeding to ligation, clean the amplified product using two columns of the Macherey-Nagel purification kit. Use the optional oligo-binding buffer for oligo binding. Elute each column with 50 µl of MB-grade water and pool down the product to obtain a total volume of 100 µl for each probe set.

■ **PAUSE POINT** The purified product can be frozen indefinitely at −20 °C.

Third-round probe amplification

● **TIMING 16 h**

13. Ligation. Add 10 µl of 1 mM third-round RCA primer again to 50 µl of the purified product from Step 12 to achieve a 60 µl total volume. Heat the suspension at 65 °C for 2 min and allow to cool to room temperature to promote annealing. Then add the whole contents to make the following master mix:

Component	Amount (µl)	Final concentration
Purified probe suspension with annealed primer	60.0	
10× T4 DNA ligase reaction buffer	7.0	1×
ATP (10 mM)	1.0	0.14 mM
T4 DNA Ligase (120 U/µl)	2.0	3.4 U/µl
Total	70.0	

Incubate foer 20 min at room temperature for ligation, heat inactivate for 10 min at 65 °C and cool to room temperature before proceeding to the next step.

▲ **CRITICAL STEP** Entire purified product from Step 12 can be used for scaling up the ligation; this will allow you to store ready to amplify circles for future experiments.

14. Third RCA (RCA-3). For the final RCA, where a large amount of material will be produced, transfer 25 µl of the ligation reaction from Step 13 to a 15 ml conical tube. Prepare the master mix to achieve a 2.5 ml reaction consisting of

Component	Amount (µl)	Final concentration
Ligated product from Step 13	25.0	
MB grade water	2,075.0	
10× phi29 buffer	250.0	1×
100× rBSA	25.0	1×
dNTP mix (25 mM each)	100.0	1.0 mM each
Phi29 DNA polymerase (10 U/µl)	25.0	0.1 U/µl
Total	2,500.0	

Incubate for 14 h at 30 °C, heat inactivate for 10 min at 65 °C and keep it at 4 °C until the next reaction.

▲ **CRITICAL STEP** This 2.5 ml reaction should result in viscous 'mucous-like' product, indicating sufficient probe amplification.

Final probe digestion and purification

● **TIMING 25 h**

15. Annealing of nicking primer. Add 50 µl of 1 mM nicking primer (this is the same primer used for second-round annealing) to the above reaction to achieve a total volume of 2.55 ml per probe set. Incubate for 2 min at 95°C and cool down to room temperature.
16. Final probe digestion reaction. Prepare a master mix of 5 ml and add to the above reaction to achieve a final volume of 7.55 ml with the following set up:

Component	Amount (µl)	Final concentration
Suspension from Step 15	2,550.0	
MB-grade water	4,450.0	
NEBuffer 2 (10×)	500.0	1×[a]
HindIII–HF (20 U/µl)	50.0	0.13 U/µl
Total	7,550.0	

[a]The final buffer concentration is based on a 5,000 µl reaction volume.

17. Incubate for 1 h at 37 °C for digestion to yield proto probes. Heat inactivate for 20 min at 65 °C and cool to room temperature for 15 min.
    ▲ **CRITICAL STEP** The reaction after this step should appear as a cloudy suspension due to the presence of a large number of denatured proteins.
18. Purification of proto probes. For the removal of precipitated proteins, incubate the solution at 95 °C for 2 min and cool the reaction on ice for 10 min to facilitate protein precipitation. Centrifuge for 10 min at 12,000*g* at 4 °C to pellet down the denatured proteins (oligonucleotides and other short DNA fragments remain dissolved in the supernatant).
19. Transfer the supernatant (~7.5 ml) to a 50 ml conical tube by pipetting slowly. Add 750 µl (~one-tenth volume) of 3 M Na–acetate to the supernatant followed by 18.8 ml (2.5 vol) of absolute ethanol. Mix by inverting the tube five to six times and place in a −20 °C freezer for at least 2 h to facilitate DNA precipitation. Centrifuge the tube at 12,000*g* for 10 min at 4 °C. Discard the supernatant by slowly decanting the tube and resuspend the pellet in 1 ml of ice-cold 70% (vol/vol) aqueous ethanol solution. Transfer the whole contents into a 1.5 ml microcentrifuge tube and centrifuge for 20 min at 12,000*g* at 4 °C. Aspirate the supernatant and leave the pellet to dry at room temperature for 5 min. Resuspend the probes in 200 µl of MB-grade water (or in buffer EB).
    ▲ **CRITICAL STEP** There will be primers and short digestion products in the probe solution that must be removed to obtain the purified probes.
    ■ **PAUSE POINT** The purified product can be stored at −20 °C for overnight.
20. To remove nicking primers and other short products, run a 15% (wt/vol) Invitrogen Novex TBE–urea gel and extract the desired probes using gel cut and crush and soak method.

To load a large volume (up to 500 µl per gel), carefully remove the precasted gel well partitions with a cut comb. Generally, precast gels are supplied with plastic combs, and the easiest way to remove the partitions is by breaking the comb to one-third of its original size and sliding the cut comb through the gel from one corner to another. Resuspend the sample with equal volume of 2× Novex TBE–urea sample buffer. Heat at 95 °C for 2 min and cool to room temperature. Carefully load the entire sample to the merged wells. Run the gel for 50 min at 180 V in 1× TBE buffer.

21. After the gel run finishes, carefully place the gel in a square Petri dish with 50 ml TBE running buffer and 5 µl of SYBR Gold nucleic acid stain. Stain the gel for 5 min and visualize on a blue-light box (with orange filter lid). Cut out the band of the desired product length with a razor and place the cut gel band into a 15 ml conical tube. Crush the band with a 3 mm glass rod or equivalent. Add 2 ml (~4 gel volumes) of buffer EB (Qiagen EB) and let the sample incubate overnight (~14 h) at 37 °C with shaking at 200 rpm.
22. Centrifuge the sample at 6,000*g* for 5 min at room temperature to pellet the gel and place supernatant (with diffused DNA probes) into a clean 15 ml conical tube. Add 200 µl of 3 M Na–acetate and 5 ml of absolute ethanol to the solution. Place the tube in a −20 °C freezer for at least 2 h to facilitate DNA precipitation.
23. Centrifuge the precipitated DNA for 10 min at 12,000*g* at 4 °C.
24. Discard the supernatant and wash the pellet by adding 1 ml 70% (vol/vol) ethanol. Centrifuge at 12,000*g* for 10 min at 4 °C and slowly decant the supernatant. Allow the pellet to air dry for 2 min and resuspend the pellet containing purified DNA probes with 500 µl EB. Measure the concentration of the protoprobes with a Nanodrop spectrophotometer using 2 µl of sample.
    ■ **PAUSE POINT** The purified protoprobes can be stored at −20 °C overnight.

UMI and poly-A addition

● **TIMING 18 h**

25. Aliquot 100 µl of protoprobes and set up a reaction for UMI and poly-A addition using a blocked extension primer (Supplementary Table 2) that contains a blocked 3′ end and a 12 bp variable nucleotide sequence flanking with a stretch of 30 thymidine residues.

Component	Amount (µl)	Final concentration
Purified protoprobes	100.0	
Extension primer (10 µg/µl)	100.0	0.4 µg/µl
NEBuffer 2.0	250.0	1×
dNTP mix (2.5 mM each)	60.0	0.06 mM
DNA polymerase I, large (Klenow) fragment (5 U/µl)	20.0	0.04 U/µl
MB-grade water	1,970.0	
Total	2,500.0	

Incubate at room temperature for 30 min and heat inactivate at 65 °C for 20 min.

26. Purify the final oligonucleotide library following Steps 20–24 and resuspend in 200 µl of EB. Measure the concentration using a Nanodrop spectrophotometer. The probes are now ready for use in single-cell experiments and are stable for at least 1 year at −20 °C.

Cell fixation

● **TIMING 1.5 h**

27. Grow bacteria in liquid media to desired OD_{600} depending on the set up of the test condition, collect and fix immediately to reserve the transcriptomes. Add 67 µl of 16% (vol/vol) formaldehyde solution to 1 ml culture for fixation and incubate at room temperature in the biosafety cabinet for 30 min.
    ▲ **CRITICAL STEP** This step is dependent upon the purpose of the single-cell experiment. Immediately fixed bacterial cultures from agar plates, infected tissues and infected cell lines can also be used. When using bacteria grown on solid surfaces, a 1% (vol/vol) solution

of paraformaldehyde in PBS can be directly applied to the bacterial sample. When using bacteria from environmental conditions (aggregates, biofilms, colonies and infected tissue culture), separation of bacterial cells from aggregates and removal of debris is required to prepare single-cell solutions, reduce background noise and accurately estimate the number of bacteria in a sample.

28. Centrifuge at 6,000*g* for 5 min at room temperature and resuspend the pellet in 1 ml of 0.02× saline–sodium citrate buffer. Centrifuge bacterial cells at 6,000*g* for 3 min at room temperature and resuspend in 350 µl of MAAM. Mix thoroughly to obtain a homogeneous cell suspension.
▲ **CRITICAL STEP** Appearance of a thick white precipitate can be observed after addition of MAAM due to the clumping of cells. This is common with many bacterial species and does not interfere with the quality of scRNA-seq data, and the cells will dissociate in later steps.
■ **PAUSE POINT** Fixed cells in MAAM can be stored at −20°C for up to 1 week without loss in RNA hybridization efficiency.

Cell permeabilization and probe hybridization

● **TIMING 16 h**

29. Preheat the thermomixer to 50 °C. Transfer ~50 µl of fixed cells from the previous step to a clean 1.5 ml microcentrifuge tube and centrifuge at 6,000*g* for 3 min at room temperature. It is recommended to optimize this step with a pellet size ~2–3 mm in diameter (~1–2 × 10^7 cells).
▲ **CRITICAL STEP** Too many cells may result in inadequate probe concentrations and result in lower than desired transcript detection, resulting in lower-quality scRNA-seq data.
30. Discard the MAAM supernatant and resuspend the pellet in 200 µl PBS for washing. Centrifuge at 6,000*g* for 3 min at room temperature and discard the supernatant. Repeat the washing step another two times and resuspend the pellet again with 300 µl PBS.
▲ **CRITICAL STEP** Carefully pipetting the cells several times is necessary to dissociate cell aggregates and to form a homogeneous cell suspension.
31. For most Gram-negative and Gram-positive bacteria, add 5.1 µl of Ready-Lyse lysozyme solution from a stock concentration of 20,560 U/µl to achieve a final concentration of 350 U/µl. Incubate at room temperature for 30 min.
▲ **CRITICAL STEP** For certain strains, optimization of cell wall permeabilization may be required. We found that for certain *S. aureus* strains, an addition of 10 µg/ml lysostaphin (Sigma, L9043) in addition to Ready-Lyse in this step will achieve robust cell permeabilization. For *Bacillus cereus* strains, addition of 400 U/ml mutanolysin (Sigma, M9901) in addition to the Ready-Lyse in this step increases probe penetration.
32. Centrifuge at 6,000*g* for 3 min at room temperature and remove the supernatant. Wash the cells by adding 300 µl of 1× PBS with 0.1% (vol/vol) Tween and pipette gently to resuspend cells. Centrifuge at 6,000*g* for 3 min at room temperature and remove the supernatant.
33. Resuspend the cells with 200 µl of room temperature HCR probe hybridization buffer and prehybridize for 15 min on a 50 °C heat block.
34. While the cells incubate, prepare the final probe library from Step 26 by heat melting and warming for hybridization. If the probes from −20 °C stock have been used, take out the probes and keep at room temperature to melt. Make an aliquot of probes depending upon the stock concentration of the probes. It is recommended to use a final concentration of 80 ng/µl in the prehybridized suspension of cells. Heat the probes at 96 °C for 2 min on a heat block and cool to room temperature on the bench. Incubate probes at 50 °C until ready to add to cells.
▲ **CRITICAL STEP** The amount of probe to be used is dependent on the probe concentration and the number of probes in the library. A final concentration of 80 ng/µl is sufficient for a library with a probe set containing 30,000 probes.
35. Once cells and probes reach 50 °C, add probe solution to cells, mix gently and incubate overnight (14–16 h) at 50 °C with shaking at 300 rpm.

Probe washes and cell enumeration

● **TIMING 2 h**

36. Aliquot HCR probe wash buffer into microcentrifuge tubes and prewarm at 50 °C on a heat block. Add 1 ml of preheated HCR probe wash buffer to the overnight incubated cell–probe suspension from Step 35. Centrifuge at 6,000*g* for 1.5 min. Carefully aspirate the supernatant and resuspend the pellet in 500 µl of fresh prewarmed wash buffer then incubate for 5 min at 50 °C with shaking at 300 rpm using a thermal mixer. Repeat the wash step for a total of seven washes with 500 µl HCR probe wash buffer, keeping the samples on the shaking heat block set at 50 °C for each wash to allow probe diffusion and to minimize the temperature variation. After these seven washes with probe wash buffer, perform two additional washes with 500 µl room temperature PBS to remove traces of wash buffer.
▲ **CRITICAL STEP** Washes with the wash buffer can be minimized to five times if a pellet is not visible after the fifth wash. In this case, proceed to the PBS washes after five cycles. Care should be taken while pipetting to avoid excessive cell loss. Gentle pipetting is recommended because shear pressure during pipetting might rupture cells upon vigorous pipetting.
37. Resuspend the pellet with 200 µl of PBS and mix gently by pipetting. The cells are now ready for counting in flow cytometer.
38. Dilute SYTO-9 green florescent nucleic acid stain (5 mM) to 1:600 in PBS to make a stock concentration of 8.3 µM.
39. To an empty well of a 96-well microtiter plate, pipette 200 µl PBS and 1 µl of SYTO-9 stain to set a blank. To count cells for each sample, load 180 µl PBS per well in the 96-well microtiter plate and use 20 µl from the cell suspension from Step 37 to make a 1:10 dilution. Mix gently and add 2 µl of SYTO-9 from Step 38 to make a final concentration of 83 nM.
40. Run the samples in a Guava easyCyte HT flow cytometer. Set the flow rate to low and run for 5,000 events with a time limit of 200 s. Estimate cells using the absolute count of FITC population. Consider the plot with FSC-H (*x* axis) versus SSC-H (*y* axis) for the size of the particles in the sample. In our system, FSC gain and SSC gain for typical bacterial cells (*B. subtilis* grown in minimal media) are routinely ~800 (Supplementary Fig. 1). Also, set the FITC-H (*x* axis) versus SSC-H (*y* axis) as appropriate for the stained whole cells. We typically set threshold for FITC to >500 and select 5,000–10,000 ungated events.
41. Use the absolute count of the FITC (right-shifted) population. The Guava can use the flow rate to calculate the number of cells/µl in the 1:10 diluted sample. Multiply the number by ten to get the cells/µl in the stock.
▲ **CRITICAL STEP** In lieu of access to a flow cytometer or to calibrate the reads from the cytometer, it is possible to count cells using a bacterial cell-sized counting chamber. To this end, we have successfully used the Petroff–Hausser counting chamber (Hausser scientific) according to the manufacturer's protocol.

10x Microfluidics GEM generation and barcoding

● **TIMING 2 h**

42. Keep cell suspension on ice and gather all the reagents. Thaw, vortex and spin down Reducing Agent B from 10x Genomics supplies.
▲ **CRITICAL STEP** Check there is no precipitate visible. If seen, vortex for 1 min to dissolve the precipitate completely.
43. Prepare PCR-1 master mix as follows:

Component	Amount (µl)	Final concentration
4× ddPCR Multiplex Supermix	19.0	1×
InDrop reverse primer (10 µM)	3.0	0.4 µM
Reducing agent B	2.0	
16,500 cells in 1× PBS	51.0	Target cell recovery 10,000
Total	75.0	

The microfluidic system encapsulates ~60% of the aqueous solution so target cell recovery can be scaled by changing the total number of cells loaded in the 51 µl of PBS, for example, to encapsulate 2,000 cells suspend 3,300 probed-cells in a total of 51 µl PBS.

▲ **CRITICAL STEP** If the cells tend to form aggregates in solution 0.1% (wt/vol) BSA (in PBS) instead of PBS can be added to prevent cell clumping.

◆ **TROUBLESHOOTING**

44. Unseal a chromium next GEM chip G microfluidic device and open the lid of the chromium next GEM holder. Assemble the chip on the holder aligning the notch found on upper-left corner of the chip and holder (Fig. 3b). Insert the chip placing left-hand side to the holder following the guide mark and press the right-hand side of the chip until a click sound is heard. Close the lid and load the samples.
45. Load 70 µl of PCR-1 master mix with cell suspension from Step 43 to the first chip well marked as well 1 using a slow pipetting action so as to not introduce air bubbles to the well (Fig. 3d). Take out the gel bead strip from the freezer and vortex with a 10x vortex adaptor for 30 s. Then bring out the gel bead strip from the holder and spin for 5 s to remove bubbles. Slowly aspirate 50 µl of gel beads by puncturing the foil seal and load into the first chip well marked as 2. Load 45 µl of partitioning oil into the first well marked as 3. Leave the bottom wells empty marked as 'no fill'. All unused wells in the rows marked as 1, 2 and 3 must be filled with 50% (vol/vol) glycerol with the corresponding volume (70 µl in well 1; 50 µl in well 2; 45 µl in well 3). Attach a 10x gasket placing shiny surface towards bottom and aligning the notch with the chip holder (Fig. 3e).

    ▲ **CRITICAL STEP** Addition of 50% (vol/vol) glycerol to unused wells is critical to maintaining the desired pneumatic pressure in all wells, and failure to fill these wells will result in run failure and can damage the 10x chromium controller. Once the samples are loaded handle the microfluidic chip horizontally to avoid wetting the gasket and do not press the gasket downwards.
46. Hold the power button of 10x chromium controller for 5 s to start up. Press the eject button to open the tray. Place the holder carefully on the tray aligning the notch. Again, press the eject button on the screen to close the tray. After the tray is inserted and the chip is recognized press the play button to run GEM generation step, initiating the encapsulation run, which lasts ~18 min. Prepare a PCR tube strip on ice to collect the GEM after running ends.
47. After the run concludes, press the eject button to open the tray and recover the microfluidic device. Take out the chip holder and remove the gasket. Open the lid of the chip holder and fold it back until a click sound is heard and the wells are positioned at a 45° angle (Fig. 3f).

    ▲ **CRITICAL STEP** Check the volume in rows labeled 1 and 2. Abnormally high left-over volume in any well may indicate a clog in emulsion formation.
48. Take a P200 pipette and slowly aspirate the content from the well labeled 3, touching the lowest point of the well. Slowly dispense the GEM into the PCR strip kept on ice from Step 46 over the course of about 20 s and touching the tip to the side wall of the tube.

    ▲ **CRITICAL STEP** GEM should appear opaque and uniform throughout the tip. Appearance of a clear proportion inside the tip containing partitioning oil indicates a potential clog.
49. Run PCR-1 inside the GEM droplets by placing the strip in a thermal cycler programmed as follows:

Step	Temperature (°C)	Time
1	94.0	5:00
2	94.0	0:30
3	50.0	0:30
4	65.0	0:30
5	Go to step 2	5 times
6	12.0	Hold

Post-GEM cleanup and cDNA amplification

● **TIMING 2.5 h**

50. Add 125 µl of recovery agent to the sample in the PCR strip at room temperature and wait for 2 min to form a biphasic mixture separating the aqueous phase containing DNA with the pink recovery agent/partitioning oil (Fig. 3g).
    ◆ **TROUBLESHOOTING**
51. Remove and discard 120 µl of the pink phase in the bottom containing recovery agent by slowly inserting the tip through the upper aqueous phase and leaving a little pink solution at the bottom (Fig. 3h).
52. Purify DNA with Macherey-Nagel PCR clean-up kit as per the manufacturer's instructions. Briefly, mix the aqueous solution from Step 51 with 200 µl binding buffer supplied in the kit. Run through the column, wash and elute with 35 µl of EB.
53. Run PCR-2 with the purified product to amplify the library. Prepare a master mix as follows:

Reagent	Volume (µl)
Purified DNA from previous step	17.0
Q5 Hot Start Master mix (2×)	20.0
10× Forward primer (10 µM)	1.5
InDrop reverse primer (10 µM)	1.5
Total	40.0

    ▲ **CRITICAL STEP** Store the remaining purified product from PCR-1 at −20 °C.
54. Run the PCR on a thermal cycler with the following cycling conditions:

Step	Temperature (°C)	Time
1	98.0	0:30
2	98.0	0:10
3	62.0	0:20
4	72.0	0:20
5	Go to step 2	15 times
6	72.0	2:00
7	4.0	Hold

55. Purify the DNA with Macherey-Nagel PCR clean-up kit according to Step 52. Elute with 30 µl of the EB. Run 5 µl of this purified DNA on a single-tier Lonza gel (1.2% (wt/vol) agarose gel) along with a 50 bp molecular size ladder. Mix 5 µl of DNA sample with 1 µl of 6× gel loading buffer and load into the well. Similarly, mix 2 µl of ladder with 1 µl of gel loading buffer to add into another well. Run the gel at 200 V for 20 min. Visualize bands in the gel directly with the blue-light box. Since Lonza gels are prestained, there is no need for staining after the gel run.
    ▲ **CRITICAL STEP** If you are not using a precast gel, make sure to add the stain in the gel while preparing or stain the gel after running has been completed. Bands should appear in the range ~200 bp if your designed probes are ~120 bp in length.
    ■ **PAUSE POINT** The purified product can be frozen till the next day.

3′ indexed Illumina library construction and data analysis

● **TIMING 3 h**

56. Run PCR-3 with the primers containing Illumina adaptors. Prepare a reaction mixture as follows:

Reagent	Volume (µl)
Purified DNA from previous step	1.0
Q5 Hot Start Master mix (2×)	25.0
10× 1-step forward primer (10 µM)	2.5
10× PCR-3 reverse primer (10 µM)	2.5
MB-grade water	19.0
Total	50.0

▲ **CRITICAL STEP** If multiple samples will be multiplexed into one pooled library that will be sequenced as a single Illumina sequencing run, use 10× PCR-3 reverse primers with unique index sequences for each sample. See Supplementary Table 2 for indexing primer sequences (10× PCR-3 R1–R16 for multiplexing of up to 16 samples).

◆ **TROUBLESHOOTING**

57. Run PCR-3 using the following cycling conditions:

Step	Temperature (°C)	Time
1	98.0	0:30
2	98.0	0:10
3	64.0	0:20
4	72.0	0:20
5	Go to step 2	5 times
6	72.0	2:00
7	4.0	Hold

58. Purify the DNA with a PCR clean-up kit (Macherey-Nagel) as per the manufacturer's instructions. Elute with 30 µl of EB. Load 5 µl of this purified DNA on a single-tier Lonza gel (1.2% (wt/vol) agarose gel) along with a 50 bp molecular size ladder. Mix 5 µl of DNA sample with 1 µl of 6× gel loading buffer and load into the well. Similarly, mix 2 µl of ladder with 1 µl of gel loading buffer to add into another well. Run the gel at 200 V for ×20 min. Visualize the bands in the gel directly with the blue-light box.

◆ **TROUBLESHOOTING**

59. Run purified DNA on Agilent D5000 TapeStation following the manufacturer's protocol to confirm presence of a band of 200–250 bp, indicating successful library generation. Briefly, add 1 µl of sample with 10 µl of sample loading buffer into the second well of a tube strip. Similarly, prepare the ladder using 1 µl of D5000 ladder with 10 µl of sample buffer and load into the first well of the tube strip. Results should indicate a sharp band at the desired size and with a concentration of ≥3 ng/µl (Supplementary Fig. 2).

▲ **CRITICAL STEP** If you are planning to multiplex several samples with different indexes and similar size of bands, you can prepare an equimolar mixture of all the samples by pooling together with equal amounts of ~20 ng each. Run the pooled library once again on TapeStation to see the proper banding position and to measure the concentration.

60. Measure the concentration of the probe library for both the type of samples, whether in single or multiplexed, using a Qubit Nanodrop spectrophotometer to recheck the concentration determined from the TapeStation together with the $A_{260/280}$ and $A_{260/230}$ to check the purity of the product. Prepare a 20–50 µl aliquot of the final library with a minimum concentration of 3 ng/µl and send it to your Illumina sequencing provider. We recommend sequencing depth of ~10,000–20,000 reads per cell but have seen good clustering and single-cell heterogeneity with sequencing depth as low as 5,000 reads per cell.

61. Once sequencing results are available as FASTQ files for each sample, align the sequences to the probe sequences using the built in Cell Ranger commands and algorithms. For this process, a probe-based reference for gene sequences must first be made. Scripts for compiling the reference 'probe-genome', .fasta and .gtf files are available on the Hormoz lab git page (https://gitlab.com/hormozlab/probacseq-toolkit). Convert these files into a 10x Cell Ranger genome using the command 'mkref' found in 10x Genomics support (https://www.10xgenomics.com/support/software/cell-ranger/latest/analysis) following 10x Genomics instructions. Then use the 'cellranger count' command to align the FASTQ reads to the genome, identify single cells, normalize the signal among the cells and provide default clustering and differential gene analysis including the Uniform Manifold Approximation and Projection (UMAP) files presented here (Supplementary Figs. 4–6).

Troubleshooting

Troubleshooting advice can be found in Table 1.

Table 1 | Troubleshooting table

Step	Problem	Possible reason	Solution
9	Viscous solution is not observed	Enzyme not sufficient or does not have high processivity	Increase the enzyme concentration or the incubation time
11	Absence of a less viscous solution	Inappropriate digestion of the product	Increase the enzyme concentration or the incubation time
43	Cell counts too low	Higher wash number or overly aggressive washings; cell lysis due to nonoptimized cell permeabilization	It is not recommended to proceed with cell suspensions of <1,000 or >10,000 total cells
50	Biphasic separation is not visible or a cloudy nonhomogeneous upper layer is observed (Supplementary Fig. 3)	Recovery agent is not distributed uniformly	Mix by inverting the capped strip five to six times, pulse spin and keep for 2 min minutes to appear the biphasic mixture
56	Faint band (low yield)	Template DNA concentration is lower than expected	Run PCR-3 again increasing the amount of template
58	Appearance of multiple bands	Nonspecific amplification can occur in high template concentration	Repeat PCR-3 with reduced amount of DNA template
58	A strong band of primer-dimers along with the desired band	Primers share a high base complementarity region	Purify desired library using DNA binding magnetic beads SPRIselect or AMPure XP, column based Select-a-size purification kit or by running a 1.5% (wt/vol) agarose gel and cutting the desired band out followed by gel purification

Timing

Steps 1–26, probe preparation: 5 d
Step 1, phosphorylation of the oligonucleotide pool: 1 h
Steps 2–7, first-round probe amplification: 17 h
Steps 8–12, second-round probe amplification: 18 h
Steps 13–14, third-round probe amplification: 16 h
Steps 15–24, final probe digestion and purification: 25 h
Steps 25–26, UMI and poly-A addition: 18 h
Steps 27–28, cell fixation: 1.5 h
Steps 29–35, cell permeabilization and probe hybridization: 16 h
Steps 36–41, probe washes and cell enumeration: 2 h
Steps 42–49, 10x Microfluidics GEM generation and barcoding: 2 h
Steps 50–55, post-GEM cleanup and cDNA amplification: 2.5 h
Steps 56–61, 3′ indexed Illumina library construction and data analysis: 3 h
(May vary depending on the time required for sequencing.)

Anticipated results

Using this protocol, bacterial single-cell transcriptomics data with high mRNA coverage per cell can be readily acquired from thousands of cells. The different subpopulations of cells expressing genes for physiological functions (motility, sporulation, competence, attachment and secretion systems) and metabolism can be observed even in well-mixed samples of clonal bacteria. For example, if applying this protocol on the classical Gram-positive model system *B. subtilis* cultured on minimal media, cells corresponding to the phenotypic attributes, such as

genetic competence, sporulation and metabolism, can be readily observed in the clusters of subpopulations as reported in literature[19] (Supplementary Fig. 4). Reliable single-cell transcriptomics data can also be achieved with anaerobic cultures as demonstrated for the pathogen *Clostridium perfringens*, in which a subset of cells in the population of the anaerobic culture has differentially expressed a toxin in the presence of acetate in the growth medium[19] (Supplementary Fig. 5).

Despite the proper design of probes, efficient encapsulation with 10x supplies and use of high-efficiency NextGen-Illumina sequencing, spurious data can also be obtained in some instances due to improper cell enumeration, malfunctioning of channels of the 10x microfluidic device resulting in a potential clog formation or inappropriate distribution of cells in GEM droplets. For example, the overloading of *B. subtilis* cells in the microfluidic device resulted in the formation of clusters probably stemming from the droplets containing multiple cells (Supplementary Fig. 6a). Another example includes droplet failure where the droplets coalesce (wetting error) and form mega-bubbles, which is indicative of forming GEMs with multiple beads and cells in a single droplet and is considered as poor-quality data in scRNA-seq experiments with no biological significance (Supplementary Fig. 6b). In contrast, other single-cell experiments using this protocol with *B. subtilis*, which have been exposed to a common DNA damaging agent for 30 min and 60 min (Supplementary Fig. 6c,d, respectively), showed distinguishable cell clusters that can be attributed to the subpopulations interacting differentially with the DNA damaging agent and are as examples of good scRNA-seq data.

Data availability

The main data discussed in this protocol are available in the supporting primary research paper[19]. The raw datasets are available for research purposes upon reasonable request from the corresponding author.

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Author contributions
A.R. and P.S. conceived the manuscript. R.M., S.F.C., P.S. and A.R. performed the laboratory experiments. S.H. and R.M. created the data analysis pipeline. P.S. and A.R. wrote the paper with input from all other authors. A.R. and S.H. supervised the project.

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The authors declare no competing interests.

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Correspondence and requests for materials should be addressed to Adam Rosenthal.

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[1]Department of Microbiology and Immunology, University of North Carolina, Chapel Hill, NC, USA. [2]Lewis-Sigler Institute for Integrative Genomics, Princeton University, Princeton, NJ, USA. [3]Department of Data Science, Dana-Farber Cancer Institute, Boston, MA, USA. [4]Department of Systems Biology, Harvard Medical School, Boston, MA, USA. [5]Broad Institute of MIT and Harvard, Cambridge, MA, USA.